# Multi-sensor Fusion  Simultaneous Localization and Mapping: A Systematic Review

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**Research Article**

**Keywords:** Multi-sensor, LiDAR, Camera, IMU, SLAM

**Posted Date:** September 13th, 2024

**DOI:** https://doi.org/10.21203/rs.3.rs-4922197/v1



**Additional Declarations:** No competing interests reported.

# Multi-sensor Fusion Simultaneous Localization and Mapping: A Systematic Review

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**Abstract**

Simultaneous Localization and Mapping (SLAM) is a crucial technology for intelligent unnamed systems to estimate their motion and reconstruct unknown environments. However, the SLAM system with merely one sensor has poor robustness and stability due to the defects in the sensor itself. Recent studies have demonstrated that SLAM systems with multiple sensors, mainly consisting of LiDAR, camera, and IMU, achieve better performance due to the mutual compensation of different sensors. This paper investigates recent progress on multi-sensor fusion SLAM. The review includes a systematic analysis of the advantages and disadvantages of different sensors and the imperative of multi-sensor solutions. It categorizes multi-sensor fusion SLAM systems into four main types by the fused sensors: LiDAR-IMU SLAM, Camera-IMU SLAM, LiDAR-Camera SLAM, and LiDAR-IMU-Camera SLAM, with detailed analysis and discussions of their pipelines and principles. Meanwhile, the paper surveys commonly used datasets and compares the effects of different algorithms. Finally, it concludes with a summary of the existing challenges and future opportunities for multi-sensor fusion SLAM.

**Keywords:** Multi-sensor, LiDAR, Camera, IMU, SLAM

# 1 Introduction

Simultaneous localization and mapping (SLAM), which aims to obtain precise locations and reconstruct the unfamiliar environment, is a foundational technique for robots to perform high-level tasks, ranging from autonomous scenario exploration to emergency rescue operations. The most commonly used sensors in SLAM systems are Light Detection and Ranging (LiDAR) and camera, corresponding to two mainstream SLAM systems, i.e., LiDAR-based SLAM (L-SLAM) (Zhang and Singh, 2014) and Vision-based SLAM (V-SLAM) (Mur-Artal et al, 2015).

L-SLAM relies on the intuitive ranging and point clouds acquisition capabilities of LiDAR to determine poses, which mainly through Iterative Closest Point (ICP) (Umeyama, 1991) and Normal Distributions Transform (NDT) (Biber and Strasser, 2003). It mainly reconstructs the surroundings by stacking point clouds based on the estimated trajectories. LOAM and its variants (Zhang and Singh, 2014; Shan and Englot, 2018) are representative works of L-SLAM. Although L-SLAM has been widely used in academia and industry, it contains certain restrictions. For instance, LiDAR measurements are always sparse and lack semantic information. In addition, L-SLAM is prone to fail in environments with degenerate geometries (eg. long tunnels, wide open spaces) and high dynamics (eg. moving obstacles and aggressive self-motion) (Tagliabue et al, 2021).

V-SLAM, which conducts SLAM based on visual sensors, estimates poses by multi-view geometry and recovers the environment by either simply stacking or implementing a full structure-from-motion (SFM) (Zhu et al, 2022b). Since cameras can provide ample data, such as dense color information and fine-grained texture, researchers have developed many V-SLAM approaches, varying from feature-based ORB-SLAM (Mur-Artal et al, 2015; Mur-Artal and Tardós, 2017) to direct method-based DSO (Engel et al, 2018) and semi-direct SVO (Forster et al, 2014). However, cameras are ambiguous in depth sensing, which hinders initialization and scale calculations. Moreover, visual sensors struggle in feature extraction with low-lighting, textureless, and high-occlusion conditions. These fatal flaws limit the adoption and accuracy of V-SLAM (Fuentes-Pacheco et al, 2015).

On the other hand, the Inertial Measurement Unit (IMU) as an environment-independent sensor, directly collects motion data of robots via accelerometers and gyroscopes at a high frequency. It is different from external sensing devices. It can offer accurate pose transformation in a short period but suffers from long-term error drifts, which require supplemental corrections from other sensors (Yang and Huang, 2018).

Obviously, these sensors can compensate each other to some extent. LiDAR and IMU can help cameras overcome scale ambiguity and address low-light conditions. IMU can support the measurement if LiDAR or cameras fail due to challenging environments, with its integral drift corrected by LiDAR and cameras. Besides, cameras can enrich the environmental information of LiDAR measurements. Moreover, intelligent unmanned systems are normally equipped with multiple sensors but each sensor serves separate tasks, resulting in insufficient and inefficient exploitation of their complementary properties. Hence, it is necessary to maximize the benefits of each sensor by multi-sensor fusion SLAM.

Current multi-sensor fusion SLAM systems are primarily based on the above-described sensors. Thus, we mainly focus on systems supported by LiDAR, camera, and IMU. They can be categorized by fused sensors into four types: LiDAR-IMU-based SLAM (LI-SLAM) (Geneva et al, 2018; Ye et al, 2019; Palieri et al, 2021; Li et al, 2021b; Xu et al, 2022a), Camera-IMU-based SLAM (CI-SLAM) (Qin et al, 2018; Huai and Huang, 2018; Yang and Huang, 2019; Campos et al, 2020; Yu et al, 2021; Stumberg and Cremers, 2022), LiDAR-Camera-based SLAM (LC-SLAM) (Huang et al, 2020; Zhang et al, 2017; Graeter et al, 2018; Shin et al, 2018; Young et al, 2020), and LiDAR-IMU-Camera-based SLAM (LIC-SLAM) (Zuo et al, 2019, 2020; Jia et al, 2021; Zhao et al, 2021; Lin and Zhang, 2022; Zheng et al, 2022; Lang et al, 2023).

Although multi-sensor fusion SLAM inherits the advantages of single-sensor SLAM and can facilitate more diverse application scenarios. Its practical implementation is pretty complex due to the heterogeneity of multiple sensors. To overcome this, several critical issues need to be paid attention to, including fusion methods, data preprocessing, and calibration. In addition, it also encounters challenges that disturb prevalent SLAM systems, such as dynamic and degenerative environments. The four types of multi-sensor fusion SLAM systems have similarities and differences in handling these challenges, while each of them has its problem to solve. Therefore, it is essential to recognize the exclusive issues in multi-sensor fusion SLAM and its exceptional treatments for common SLAM difficulties. We systematically summarize the essential basics of the multi-sensor fusion SLAM, as well as analyze the advantages and disadvantages of systems with different sensor fused types. Furthermore, we investigate the implementation of deep learning in multi-sensor fusion SLAM, which has not been fully explored in this field. Finally, we discuss the potential direction of the future development and improvement of multi-sensor fusion SLAM.

Several relevant works survey multi-sensor fusion SLAM. Most of them only focus on one type of sensor fusion, such as camera and IMU (Huang, 2019), camera and LiDAR (Chghaf et al, 2022), and 3D LiDAR-based combinations (Xu et al, 2022b), thus lacking a systematic analysis of the varied compositions among these three sensors. Ebadi et al. (2024) survey the extraordinary works in *DARPA Subterranean Challenge*[1], which is mainly focused on engineering practices of multi-sensor fusion SLAM for this specific competition. While some works contain multi-sensor fusion (Zhu et al, 2024), they merely concentrate on conventional algorithms and lack investigation of learning-based methodologies. We additionally survey learning-based multi-sensor fusion SLAM and group it as a fusion framework, demonstrating new possibilities for end-to-end approaches. This paper aims to provide a systematic and comprehensive overview of the multi-sensor fusion SLAM systems. It contains the infrastructural issues and analysis of various fused types of multi-sensor fusion SLAM, as well as their connections and differences.

The remaining of this paper is organized as follows: Sec.2 introduces the preliminary knowledge of multi-sensor fusion SLAM, including residuals of sensor measurements, fusion frameworks, and calibration. Four categories of multi-sensor fusion SLAM algorithms are reviewed and analyzed in detail in Sec.3. The popular datasets, performance evaluation, and hardware configurations are involved in Sec.4.

[1] https://www.darpa.mil/program/darpa-subterranean-challenge

Finally, challenges and promising research directions are discussed in Sec.5, and Sec.6 concludes the paper. The whole framework of this paper can be shown as Fig. 1.

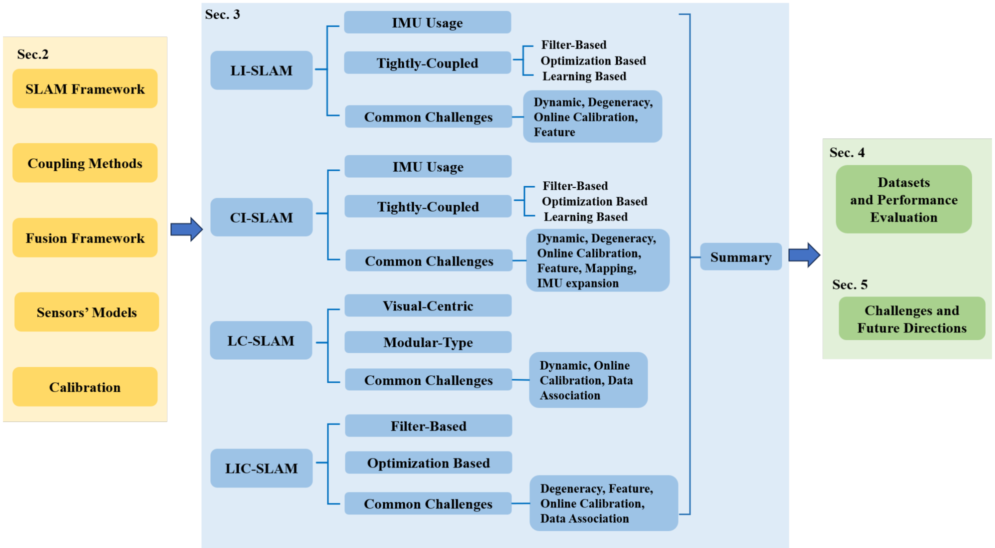


**Fig. 1** The overview of this review.

# 2 Preliminary Knowledge

In this section, we provide the basic background of multi-sensor fusion SLAM, disassembling the SLAM framework for the multi-sensor case and introducing the implementation requirements.

## 2.1 SLAM Framework

Multi-sensor fusion SLAM can be separated into *front-end* and *back-end*. The front-end extracts valid data and employs pre-processing, while the back-end infers and optimizes based on the data output by front-end. The general framework of multi-sensor fusion SLAM is shown in Fig.2.

### 2.1.1 Front-end

The front-end is composed of data preprocessing, system initialization, and data association. Data preprocessing is retrieving beneficial measurements from diverse heterogeneous data, including calibration, undistortion, and feature extraction. System initialization refers to obtaining the initial state of the system, such as scales and IMU bias. Data association contains data-level and time-level, the former addresses the spatial correspondence of the camera and LiDAR observations, while the latter detects loop closure in fresh and historical measurements.

### 2.1.2 Back-end

The back-end integrates and utilizes information across the board to acquire global consistent pose estimation and reconstruct the map of the environment. The state vector $X$ to be estimated can be shown in (1). It usually contains the position ${}^{G}p$ and rotation ${}^{G}R$ of the sensing platform, with other estimable variables involved in a multi-sensor fusion SLAM system, such as the velocity ${}^{G}v$, gravity ${}^{G}g$, and biases $b$ from IMU, a set of map points ${}^{G}P$, extrinsic matrix $T_{calib}$ between sensors, and the inverse depth $\lambda$ of $m$ image points. The upper-left corner marker {G} indicates that the variation is relative to the world coordinate system.

$$X = [{}^{G}R, {}^{G}p, {}^{G}v, {}^{G}g, b, {}^{G}P, T_{calib}, \lambda_0, ..., \lambda_m] \tag{1}$$

Intuitively, SLAM is a state estimation problem and another method of estimating trajectory is Odometry. The distinction between SLAM and Odometry is whether or not the system enables map construction. Nevertheless, the recent advent of Visual Odometry (VO) and LiDAR Odometry (LO) integrated with loop closure has gradually diffused the boundary of Odometry and SLAM (Huang, 2019). Hence, this paper also reviews multi-sensor fusion Odometry since it is a simplified SLAM.

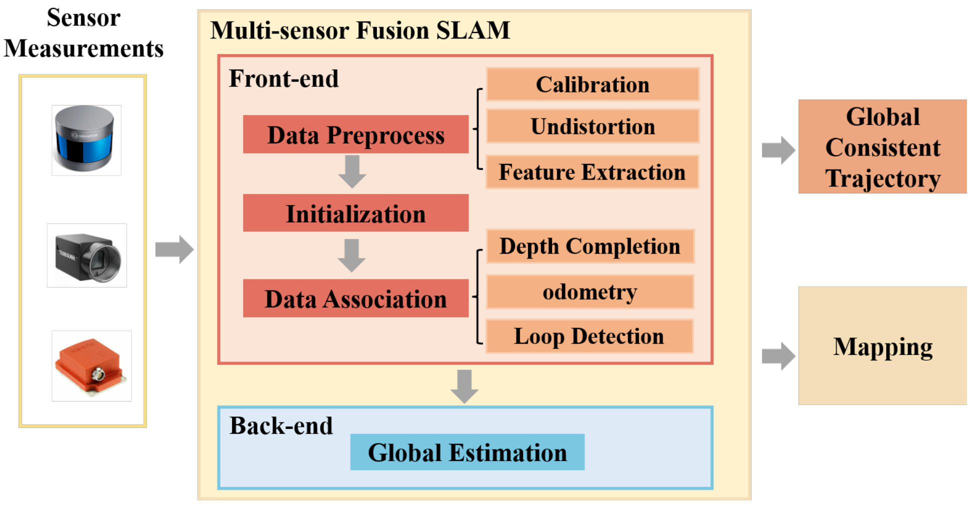


**Fig. 2** The overview of multi-sensor fusion SLAM framework. The red box represents the front-end, which contains data preprocess, initialization, and data association. The blue box denotes the back-end that provides global constraints for trajectory estimation and mapping.

## 2.2 Coupling Method

Multi-sensor fusion SLAM systems can be broadly classified as *loosely-coupled* and *tightly-coupled* based on when the measurements are fused.

### 2.2.1 Loosely-Coupled

The loosely-coupled systems handle data from different sensors individually and focus more on combining the intermediate estimations instead of raw measurements. Fusion on the intermediate estimations decreases the computation load and makes the loosely-coupled manner resilient and manageable. Hence, the loosely-coupled systems are generally applied to withstand sensor degradation or noise interference (Palieri et al, 2021; Han et al, 2023; Shao et al, 2019; Qin et al, 2020b; Beauvisage et al, 2022; Khattak et al, 2020; Khedekar et al, 2022).

Although loosely-coupled sensor fusion is appealing for runtime and robustness, it ignores correlations among diverse modalities and may cause information loss, convergence issues, and inaccurate estimation. Moreover, in extreme environments, the intermediate estimations of some sensors will fail and cause unreliable fusion in later stages.

### 2.2.2 Tightly-Coupled

The tightly-coupled approaches directly fuse measurements of sensors to avoid the decoupling of constraints. Benefiting from the capitalization on the full range of heterogeneous data, the tightly-coupled systems generally have high accuracy. They heavily depend on precise models of sensors and their noise (Ebadi et al, 2024). To achieve better performance, the majority of multi-sensor fusion SLAM systems favor the tightly-coupled approach (Zhao et al, 2019; Qin et al, 2020a; Xu and Zhang, 2021; Bai et al, 2022; Campos et al, 2021; Qin et al, 2018; Huai and Huang, 2018; Qin et al, 2019; Heo et al, 2019; Huang et al, 2020; Wisth et al, 2021; Jia et al, 2021; Shan et al, 2021; Lin and Zhang, 2022).

## 2.3 Fusion Frameworks

Coupling methods determine when the measurements are fused while fusion frameworks indicate how the messages are integrated, which can be categorized as *filter-based*, *optimization-based*, and *learning-based*. Both loosely-coupled and tightly-coupled systems can use either filter-based or optimization-based fusion frameworks, yet learning-based fusion mainly exists in tightly-coupled systems.

### 2.3.1 Filter-based

The prevailing filter-based multi-sensor fusion approaches generally adopt the Extended Kalman Filter (EKF) and its variants, such as Iterated Extended Kalman Filter (IEKF) (Barfoot, 2024). EKF consists of two steps: propagation by kinematic equation and updating by observation equation. In multi-sensor fusion SLAM systems with IMU, the kinematic equation is always formed by IMU measurements and the observation equation generally established by LiDAR or camera data.

Filter-based systems are computational and memory efficient, thus widely applied in miniaturized embedded systems (Zhao et al, 2019; Qin et al, 2020a; Xu and Zhang, 2021; Xu et al, 2022a; Bai et al, 2022; Huai and Huang, 2018; Geneva et al, 2020; Qin et al, 2020b; Van and Mahony, 2023; Wisth et al, 2021; Jia et al, 2021; Wang et al, 2019; Shan et al, 2021). However, there are several limitations. Since filter-based methods process measurements sequentially, they are sensitive to time synchronization and cannot correct past states with feature measurements (Sandy et al, 2019). In addition, they suffer from inconsistency caused by linearization error and observability mismatch. These drawbacks of filter-based methods result in worse accuracy than optimization-based methods as revealed in (Strasdat et al, 2012).

### 2.3.2 Optimization-based

Different from filter-based approaches that incrementally estimate state, optimization-based methods jointly process numerous observations through nonlinear least squares optimization. This process is also called Bundle Adjustment (BA). The formulation can be expressed as follows:

$$\mathcal{X}^* = arg\min_{\mathcal{X}} \left\{ \sum_{i \in \mathcal{I}} \|r_i\|^2_{\Sigma_I} + \sum_{l \in \mathcal{L}} \|r_l\|^2_{\Sigma_L} + \sum_{c \in \mathcal{C}} \|r_c\|^2_{\Sigma_C} + \sum_{o \in \mathcal{O}} \|r_o\|^2_{\Sigma_O} \right\} \tag{2}$$

where $\mathcal{X}$ represents a set of state vectors. $r_{(\cdot)}$ and $\Sigma_{(\cdot)}$ are respectively the residuals and covariance of different measurements. The subscripts represent sensors they belong to. $\{i\}, \{l\}, \{c\}$ correspond to IMU, LiDAR, and camera respectively, and $\{o\}$ stands for other measurements, including other sensors, *priori*, and loop closure. The decorated letters denote a set of measurements. Eq. 2 is to identify a set of state vectors that minimize individual residual terms, which will be interpreted in Sec.2.4.

Optimization-based methods are always represented by factor graph. In a factor graph, nodes represent variables to be optimized and edges indicate the constraints on the linked variables, which are formed by measurements. The edges are also called factors. Since the flexibility of the factor-graph representation and the superior accuracy of BA, optimization-based methods have been broadly adopted in multi-sensor fusion SLAM systems (Geneva et al, 2018; Ye et al, 2019; Shan et al, 2020; Le Gentil et al, 2021; Nguyen et al, 2021; Li et al, 2022; Campos et al, 2021; Mascaro et al, 2018; Qin et al, 2018; Xie et al, 2020; Seok and Lim, 2020).

However, joint optimization requires a huge amount of computation and memory. Prevalent methods alleviate this problem by tricks like sliding windows (Qin et al, 2018; Shan et al, 2020) and keyframes (Shan et al, 2020). The highly optimized solvers specifically designed for BA (e.g., G2O (Kümmerle et al, 2011), Ceres (Agarwal et al, 2022)) and incremental solvers (Kaess et al, 2008) are commonly adopted to accelerate the computation. Nevertheless, the computational complexity and memory consumption of optimization-based methods are still generally more expensive than filter-based approaches. This flaw restricts their applications only to systems without memory and time limitations.

### 2.3.3 Learning-based

Current learning-based fusion exploited in multi-sensor fusion SLAM mainly extracts different features of diverse modalities by deep neural networks (DNNs) (e.g., CNN for images and LiDAR, LSTM for IMU) and then fuse the extracted features by deep learning strategies (e.g., Transformer (Sun et al, 2023) or attention mechanism (Liu et al, 2021)). The fused features are fed into DNNs for pose inference and map construction. This learning-based fusion framework can generate an end-to-end system.

However, most learning-based multi-sensor fusion SLAM systems now are limited to Odometry (Clark et al, 2017; Han et al, 2019; Shamwell et al, 2018a; Sun et al, 2023) since mapping with DNNs is usually achieved offline (Mildenhall et al, 2021).

Whereas learning-based methods have not been applied in practices as extensively as traditional fusion frameworks and generally consume more time, they tend to achieve higher accuracy and are more robust owing to the superior capacity of DNNs in information extraction and fusion.

In summary, each fusion framework has its advantages and disadvantages, which lead to algorithmic trade-offs for different situations.

## 2.4 Residual of Different Sensors

The residuals from distinct sensors are dependent on their data collection instruments and feature definition methods.

### 2.4.1 LiDAR

LiDAR is an active sensor, whose measurements are mainly composed of the intensities and 3D coordinates of point clouds in the LiDAR frame. According to the scan patterns, they can be clarified as mechanical-based and solid-state-based. The distinction of scan patterns results in different feature definition methods. Systems with mechanical-based LiDARs (Geneva et al, 2018; Ye et al, 2019; Zhao et al, 2019; Shan et al, 2020) commonly adopt the curvature calculation proposed in LOAM (Zhang and Singh, 2014). However, it is not suitable for solid-state LiDARs. Take Livox MID40 LiDAR as an example, it has a small conical-shaped FOV in the front and adopts a non-repetitive rosette-like scanning pattern (Lin and Zhang, 2020). Lin and Zhang (2020) traversed along the incident and deflection angle to choose point candidates for Livox MID40.

The measurements of LiDAR are generally used as residuals from the scan-to-map or scan-to-scan matching. The residuals $r_l$ are expressed as point-to-line distance $r_{line}$ or point-to-plane distance $r_{plane}$.

$$\begin{aligned} r_{line} &= l \times ({}_L^G R {}^L P + {}^G t) + p_0 \\ r_{plane} &= n^T ({}_L^G R {}^L P + {}^G t) + d \end{aligned} \tag{3}$$

where ${}_L^G R$ and ${}^G t$ denote the rotation and translation of LiDAR; ${}^L P$ is a point cloud scanned in LiDAR frame; $l$ is the direction vector of the line; $p_0$ is a point on the line; $n^T$ and $d$ denote the normal vector of the plane and the distance the plane to the origin, respectively.

### 2.4.2 IMU

IMU can sequentially measure the sensor's acceleration and angular velocity in its frame. The extensively utilized model is the IMU *preintegration* and its residual (Lupton and Sukkarieh, 2012; Forster et al, 2017). The IMU preintegration is composed of measurements between two keyframes at time $t_i$ and $t_j$, which can be represented as rotation $\Delta\tilde{R}_{ij}$, velocity $\Delta\tilde{v}_{ij}$, and translation $\Delta\tilde{p}_{ij}$. Suppose the increment of bias is $\delta b$ and update the preintegration measurement using a first-order expansion of the motion model (Murray et al, 2017), the residuals of preintegration can be modeled as:

$$
\begin{aligned}
r_{\Delta R_{ij}} &= Log\left(\Delta\tilde{R}_{ij}^{T} R_i^T R_j\right) \\
r_{\Delta v_{ij}} &= R_i^T(v_j - v_i - g\Delta t_{ij}) - \Delta\tilde{v}_{ij} \\
r_{\Delta v_{ij}} &= R_i^T(p_j - p_i - v_i\Delta t - \frac{1}{2}g\Delta t_{ij}^2) - \Delta\tilde{p}_{ij}
\end{aligned}
\tag{4}
$$

where $R_i$ and $R_j$ denote the rotation of IMU relative to the global frame at $i$ and $j$ timesteps; $p_i$ and $p_j$ denote the translation; $v_i$ and $v_j$ denote the velocity; $\Delta t_{ij}$ is the time interval.

### 2.4.3 Camera

The residual of camera measurements $r_c$ can be categorized as geometric error $r_g$ and photometric error $r_p$, corresponding to feature-based and direct V-SLAM, respectively. The geometric error relies on the imaging models of cameras such as the pin-hole model and MEI mode (Mei and Rives, 2007). Express the projection function in terms of $\pi$, the reprojection error between matched 3D points $P_i^G$ in the global frame and keypoints $p_i$ with $i \in \mathcal{M}$ the set of all matches can be modeled as:

$$
r_g = \sum_{i\in\mathcal{M}} \left\| p_i - \pi\left(RP_i^G + t\right)\right\|_{\Sigma}^2 \tag{5}
$$

where $R$ and $t$ denote the rotation and translation of the camera; $\Sigma$ is the covariance matrix associated with the scale of the keypoint.

The photometric error is based on the photometric consistency assumption and thus is more sensitive to illumination change. Define $I_{(\cdot)}(p_{(\cdot)})$ as the intensity of point $P_j^G$ in different frames, the photometric constraints between the host frame $m$ and the target frame $n$ can be derived as:

$$
r_p = \sum_{j\in\mathcal{N}} \left\| I_m(p_{m,j}) - I_n(p_{n,j})\right\|^2 \tag{6}
$$

where $\mathcal{N}$ is the set of pixels included in the calculation.

## 2.5 Calibration

In this section, we mainly focus on the offline calibration toolboxes that calibrate before system operation and divide them into *spatial* and *temporal*.

### 2.5.1 Spatial Calibration

Spatial calibration acquires the extrinsic matrix between sensors. For LiDAR and IMU, *lidar_align*[2] estimates by minimizing the K-nearest neighbor error after point cloud projection. Lv et al. (2020) calculate the extrinsic through batch optimization based on the B-Spline model[3]. Zhu et al. (2022a) consider the time offset after linear interpolation and estimate it based on cross-correlation. They achieve on-the-fly performance without hardware setup or help from another sensor[4]. In terms of IMU and camera, *kalibr*[5] obtains the extrinsic by aligning the pose estimation of the two sensors. For LiDAR and camera, methods usually match their observation of the external world. Yuan et al. (2021) propose to extract edge features directly in the solid-LiDAR point cloud and then estimate the extrinsic by minimizing the distance between LiDAR edge points and their corresponding image points searched by kd-tree[6]. Besides, *Direct_visual_lidar_calibration*[7] (Koide et al, 2023) and *Autoware*[8] are available online. For a more comprehensive artifact, (Yan et al, 2022) open-source a versatile multi-sensor calibration toolbox named *OpenCalib*[9]. It can calibrate both extrinsic and intrinsic matrices of sensors.

### 2.5.2 Temporal Calibration

In the application of sensor fusion, sensors are normally triggered by a uniform clock. However, due to the sensor's data processing or delays, there may be a discrepancy between the true measurement time and its timestamp (Li and Mourikis, 2014). This leads to a constant time offset $t_d$ between different sensor sequences. Temporal calibration generally handles $t_d$ through hardware (e.g., FPGA (Wenzel et al, 2021)) or software synchronization (e.g., Network Time Protocol[10]). After unifying the clock source, measurement differences caused by frequency variations are generally handled by linear interpolation (Xu and Zhang, 2021).

Although offline calibration contains many open source toolboxes, it is inconvenient because of the need to re-operation each time the hardware changes. Besides, strict hardware synchronization is not always available for low-cost or self-assembled devices (Qin and Shen, 2018). Thus, many multi-sensor SLAM systems contain the online calibration function and we will discuss this in detail in Sec.3.

# 3 Fusion SLAM Algorithm

We mainly focus on the multi-sensor fusion SLAM algorithms in recent years and classify the multi-sensor fusion SLAM systems according to the fused sensors as LI-SLAM, CI-SLAM, LC-SLAM, and LIC-SLAM. For each type of system, we first choose the

[2] https://github.com/ethz-asl/lidar_align
[3] https://github.com/APRIL-ZJU/lidar_IMU_calib
[4] https://www.github.com/hku-mars/LiDAR_IMU_Init
[5] https://github.com/ethz-asl/kalibr
[6] https://github.com/hku-mars/livox_camera_calib
[7] https://github.com/koide3/direct_visual_lidar_calibration
[8] https://github.com/autowarefoundation/autoware
[9] https://github.com/PJLab-ADG/SensorsCalibration
[10] https://datatracker.ietf.org/doc/html/rfc5905

appropriate classification method from the coupling approach or the fusion frameworks and then analyze their solutions for common SLAM challenges and specific problems.

## 3.1 LiDAR-IMU SLAM

LI-SLAM systems originate from L-SLAM systems. One of the milestones of L-SLAM algorithms is LOAM (Zhang and Singh, 2014), which defines local curvature to extract edge and planar features. Many LI-SLAM works are variations of LOAM or employ its feature extraction manner (Qin et al, 2020a; Shan et al, 2020; Xu and Zhang, 2021; Li et al, 2021b; Lv et al, 2021; Jiang and Shen, 2022).

Since LiDAR continuously scans the environment when the system keeps moving, points in one scan are obtained at different timestamps, resulting in motion distortion. In LI-SLAM systems, a universal application of IMU is the correction of LiDAR point clouds (de-skew) in the data preprocess stage of the front-end (Zhao et al, 2019; Shan et al, 2020; Palieri et al, 2021; Xu and Zhang, 2021; Nguyen et al, 2021; Li et al, 2021c; Liu et al, 2022; Júnior et al, 2022).

The majority of methods assume that time is discrete and undistort LiDAR points by interpolations. IMU locates the corresponding point clouds and projects them into a unified LiDAR scan according to the timestamped poses measured by it, which also contributes to the dynamic object filter (Zhao et al, 2019; Shan et al, 2020). However, discrete-time-based approaches have some inherent limitations. Their interpolations are accompanied by errors. Besides, handling ample raw measurement directly is intractable. To avoid these problems, researchers apply the continuous time model (e.g. B-spline (Sommer et al, 2020)) to involve the undistortion in trajectory estimation (Le Gentil et al, 2021; Lv et al, 2021; Chen et al, 2023b). Chen et al. (2023b) design a two-step coarse-to-fine point cloud deskew method by first coarsely constructing the trajectory through IMU integration and then refining it based on continuous-time equations. It is worth mentioning that distinct from the above methods, He et al. (2023) fuse each LiDAR point at its true sampling time rather than a scan to avoid motion distortion with IMU.

Early LI-SLAM works are mainly loosely-coupled that either simply regard the transformations from IMU as priors or directly integrate odometry results (Zhang and Singh, 2014; Zhen et al, 2017). Nowadays loosely-coupled LI-SLAM approaches are more delicate in utilizing the resilience of the loosely-coupled manner, which enables modality change in specific conditions (Palieri et al, 2021; Reinke et al, 2022; Han et al, 2023).

To comprehensively use information from all sensors, popular LI-SLAM systems are predominantly tightly-coupled and can be categorized as *filter-based*,*optimization-based*, and *learning-based* ones according to the fusion framework.

### 3.1.1 Filter-based

The main drawback of filter-based approaches is the cumulative error since EKF is susceptible to linearization errors (Huang and Dissanayake, 2007). General improvements attach importance to modifying the EKF, such as IEKF and iterated error-state Kalman Filter (IESKF). LINS (Qin et al, 2020a) presents a robot-centric IESKF, which recursively corrects the estimated state by generating new features in each iteration. Xu and Zhang (2021) exploit IESKF to reduce linearization errors and propose a new complexity-reduced formula to calculate the Kalman gain. Fasterlio (Bai et al, 2022) applies the Anderson acceleration to the iteration of IESKF (AA-IESKF) (Gao et al, 2022).

In addition to fixing the cumulative error, many works also focus on enhancing the computational efficiency of the filter-based LI-SLAM systems. The computation speed can be accelerated by optimizing the data structure of point clouds (Xu et al, 2022a; Bai et al, 2022; Reinke et al, 2022; Jiang and Shen, 2022; Ji et al, 2024). The classical k-d tree structure requires time to build, iterate, remove unwanted nodes, and reconstruct the tree, which influences the efficiency. FAST-LIO2 (Xu et al, 2022a) develops incremental kdtree (ikdtree) (Cai et al, 2021) to represent dense point clouds. This novel data structure enables dynamic updates of the tree and reduces the computation cost of dense registration of point clouds, which makes FAST-LIO2 a particularly outstanding achievement in the LI-SLAM field. The hash-based voxel structure can further speed up the system. Bai et al. (2022) present the sparse incremental voxel (iVox) with the help of the k-d tree and Hilbert curve. Ji et al. (2024) further propose an iVox that only contains a centroid and covariance matrix. They utilize the easy addition and deletion characteristic of voxels and ease the computation burden. Some other dynamic data structures such as octree have been also adopted to manage the map (Reinke et al, 2022; Jiang and Shen, 2022).

#### 3.1.2 Optimization-based

In addition to the IMU preintegration factor and LiDAR factor, some optimization-based LI-SLAM systems include other factors to improve their performance. The widely used one is the loop detection factor which acts as a constraint when the system revisits a place (Shan et al, 2020; Liu et al, 2022). Researchers also design novel factors for specialized constraints, such as new features (Geneva et al, 2018), gravity (Wang et al (2023b)), dynamic tracking (Lin et al, 2023), and matching cost (Koide et al, 2022). Koide et al (2022) devise a voxelized Generalized Iterative Closest Point (GICP) matching cost factor, which employs more points in the calculation of registration error than the conventional line and plane point matching.

To overcome the time-consuming drawback of the optimization-based method, researchers leverage critical components of the measurements rather than using the whole data in optimization (Shan et al, 2020; Li et al, 2021b; Nguyen et al, 2021; Lv et al, 2021; Koide et al, 2022). Tan et al. (2020) present LIO-SAM, a fairly classic and versatile optimization-based LI-SLAM system that utilizes key-frame, sliding window, marginalization, and local map. It selects a keyframe when the pose change exceeds a threshold relative to the previous state. The sliding window is used to create a local point cloud map for scan-matching and the old LiDAR scans are marginalized as priors. LiLi-OM (Li et al, 2021b) adopts a hierarchical keyframe-based sliding window optimization to fuse IMU and LiDAR measurements. Besides, it restricts the optimization time to accelerate state estimation.

### 3.1.3 Learning-based

In contrast to the extensively studied conventional frameworks, there are relatively few learning-based LI-SLAM. Most of them exploit CNN for LiDAR and LSTM for IMU feature extraction (Son et al, 2021; Iwaszczuk et al, 2021; Tu and Xie, 2021). Recent methods leverage the globality and generalisability of the Transformer to fuse information of different modalities. TransFusionOdom (Sun et al, 2023) projects LiDAR and IMU measurements in the image plane, which is suitable for CNN to extract features. It fuses the homogeneous and heterogeneous information through soft mask attention and Transformer respectively, and achieves quite encouraging results on the KITTI dataset (Geiger et al, 2012).

### 3.1.4 Common Challenges

In addition to improvements tailored specially to deal with the shortcomings of fusion frameworks, LI-SLAM contains several solutions for dynamic, degeneracy, and online calibration and is often improved by enhancing feature extraction.

**Dynamic:** High dynamics issue combats the static hypothetical of SLAM systems. The environmentally independent nature of IMU relives the influence of dynamic objects to a certain extent. A more reliable method is to combine DNNs in dynamic filtering. LIOM (Zhao et al, 2019) introduces Fully Convolutional Neural Networks (FCNN) for dynamic object removal and obtains static background point clouds. Apart from learning-based approaches, dynamic points can be detected according to the visibility assumptions, i.e., if we observe the far point of the map, then the nearby points in the query scan must be moving (lio, 2021). These methods always treat dynamic points as disturbances and reject them after detected.

**Degeneracy:** Most LI-SLAM systems pay more attention to the degeneracy in LiDAR measurements caused by the environment. Several methods inherit the idea in (Zhang et al, 2016) that inspect degeneracy by the Hessian of LiDAR measurement (Shan et al, 2020; Tagliabue et al, 2021; Han et al, 2023). LION (Tagliabue et al, 2021) analyzes the Hessian of the point-to-plane ICP function and chooses another odometry by HeRO (Santamaria Navarro et al, 2019) when degeneracy is too large. Since the system runs in long tunnels, it deems the translation part to be the most challenging to estimate. Thus, its observability metric relies on the ratio between the largest and smallest eigenvalues of the translation component. The frequency of measurement is also a criterion for degeneracy. LOCUS (Palieri et al, 2021) contains a simple health monitor module that deems the input as healthy when the rate is sufficient (>1 Hz). Otherwise, it will choose another measurement according to its predefined priority queue of modalities. Some methods assess the quality and number of the detected features (Júnior et al, 2022) or devise uncertainty models for LiDAR measurements (Jiang and Shen, 2022). Jiang and Shen (2022) devise a principled uncertainty model and a noise estimation method for LiDAR measurements to mitigate the uncertainty estimation problem.

**Online Calibration** Online calibration optimize calibration variants along with state vectors. IN2LAAMA (Le Gentil et al, 2021) involves time offset in IMU preintegration and builds a time-shift residual through IMU bias. The extrinsic is utilized when extracting features and is included in the feature residuals. It integrates time offset and extrinsic into state variables and calibrates by jointly optimizing the time-shift factor, feature factor, and IMU factor.

**Feature:** Feature extraction in the front-end is one of the preliminaries for the state estimation. In terms of craft geometric descriptors for edges and planes, the gravity and angular information from IMU is extremely helpful (Geneva et al, 2018; Li et al, 2021c; Wang et al, 2023a). LIPS (Geneva et al, 2018) proposes a novel formulation using the closest point plane representation to overcome the singularity issue of Hessian and generates an anchor plane factor. Li et al. (2021c) identify the ground plane based on the gravity vector from IMU. They determine corner points by calculating the angle between the target point and two adjacent points, which later assists the aggregation of lines and surfels. As for solid-state LiDAR, Li et al. (2021b) tailor a novel feature extraction method for Livox Horizon performed in the time domain. Some approaches make extra use of the intensity measured by LiDAR. Li et al. (2022) combine intensity and geometry to find edge points that help to resist the degeneracy of solid-state LiDAR. Zhang et al. (2023d) introduce reflectivity texture information to build photometric residuals blended with geometric residuals by IEKF. Screening for key components in all features can also provide beneficial effects. KFS-LIO (Li et al, 2021d) refers to the Dilution Of Precision concept (Kaplan and Hegarty, 2017) in satellite position. It introduces a quantitative LiDAR feature evaluation module that counts the Jacobian of the state error of position for feature selection.

## 3.2 Camera-IMU SLAM

Obtaining scale information is conducive to monocular cameras with scale ambiguity. Since IMU can provide measurements of ego-motion like accelerations and attitude, the general applications of IMU in CI-SLAM lay in metric scale recovery, which is typically implemented during the initialization of the systems.

The purpose of the initialization of CI-SLAM is to retrieve essential parameters of the system, including metric scale, gravity direction, velocity, and biases of IMU. Some methods integrate scale and gravity direction into the optimization model and initialize immediately with an arbitrary scale (Von et al, 2018). More generally, with the advent of VIN-MONO (Qin et al, 2018), the initialization method it employs has become a classical paradigm (Qin and Shen, 2017; Qin et al, 2018; Seok and Lim, 2020; Campos et al, 2021; Zhang et al, 2023c). It first performs an up-to-scale visual-only SFM and then loosely aligns it with the IMU preintegration in a restricted sliding window (Qin and Shen, 2017). ORB-SLAM3 (Campos et al, 2021) improves this approach by exploiting an inertial-only initialization method which considers IMU measurement uncertainty before aligning the IMU trajectory and V-SLAM trajectory. It can achieve sophisticated initialization in less than 4 seconds (Campos et al, 2020). However, IMU is unable to initialize with the motion of constant velocity. Stumberg and Cremers (2022) deal with this problem by continuously estimating the scale and gravity direction after initialization. They present delayed marginalization and pose graph bundle adjustment (PGBA) to alleviate the computing effort and inject IMU factors into marginalized states for IMU initialization. The robust and long-term promised scale and gravity direction estimation make DM-VIO perform better than VINS-MONO.

In particular, robot-centric formulation avoids the need to align the initial orientation with the global gravitational direction and allows the CI-SLAM to start at any pose (Huai and Huang, 2018).

Stereo cameras are not limited by scale ambiguity since they can calculate depth directly. Stereo-SLAM algorithms (Mur-Artal and Tardós, 2017) can use this depth and generally performance better than pure monocular-SLAM. However, stereo cameras need much more complex calibration than monocular cameras, making them more difficult to deploy. Their depth is limited by the baseline and they will inevitably encounter the same challenges that monocular cameras do in dynamic and textureless environments (Yu et al, 2023). Stereo-inertial SLAM focuses on other perspectives rather than depth, such as reducing computing consumption (Sun et al, 2018), stereo-IMU calibration (Yin et al, 2023b), IMU-assisted feature extraction (Liu et al, 2023), unsupervised method (Han et al, 2019), and utilization of scene flow (Yin et al, 2023a; Wen et al, 2024a). Stereo cameras bring more computation burden because of the additional camera included. Sun et al (2018) alleviate this by representing the stereo observation in $\mathbb{R}^4$ to avoid the need for stereo rectification.

Analogous to LI-SLAM systems, early CI-SLAM systems are primarily loosely-coupled (Forster et al, 2015), while the tightly-coupled ones are dominantly researched today. We analyze the tightly-coupled CI-SLAM following distinct fusion frameworks.

### 3.2.1 Filter-based

Hesch et al (2014) derive that the linearization error of EKF will change the unobservable directions of CI-SLAM systems and thus cause inconsistency. The solutions of this issue focus on the linearization point management (Huang et al, 2009), observability constraint (Hesch et al, 2012), the robot-centric formulation (Bloesch et al, 2015; Huai and Huang, 2018), and the refinement of the filter (Yang et al, 2022; Van and Mahony, 2023). Modifying the EKF linearization point represented by the First-Estimates Jacobian (FEJ) method (Huang et al, 2009) ensures the null space of the observability matrix does not degenerate. OC-EKF (Hesch et al, 2012) explicitly enforces observability constraints. The robot-centric formulation in ROVIO (Bloesch et al, 2015) and R-VIO (Huai and Huang, 2018) avoids the alignment with the global gravity vector and the observability mismatch of the world-centric CI-SLAM. In addition, the invariant EKF (Bonnabel, 2007) and the equivariant filter (EqF) (van Goor et al, 2023) based on the Lie group have been adopted to circumvent the consistency issue. Van and Mahony (2023) prove that the EqF based on their newly proposed Lie group is a consistent estimator for CI-SLAM with lower linearization error and higher order output. Furthermore, the observability changes when extracted visual features change. Yang and Huang (2018, 2019); Yang et al (2019a) analyze the uncertainty and unobservable directions of ordinary visual geometric features and their proper expression. They propose two unified forms of visual feature representation: the quaternion and closet point (CP), which is later exploited for CP point and line representations in Yang et al (2019a), and derive the unobservable subspace for multiple lines and planes (Yang and Huang, 2018, 2019).

Generally, filter-based CI-SLAM systems derive from Multi-State Constraint Kalman Filter (MSCKF) (Mourikis and Roumeliotis, 2007), following the alternative process of IMU-based propagation and visual-based updates. MSCKF maintains several past cloned camera poses over a sliding window in the state vector instead of feature points and thus avoids the dimensional explosion. A wide range of filter-based CI-SLAM works inherit the framework of MSCKF and enhance it in various aspects (Brossard et al, 2018; Eckenhoff et al, 2019c,a; Heo et al, 2019; Geneva et al, 2019a; Zou et al, 2019; Chen et al, 2023a). Several methods leverage the computation efficiency and high non-linearity from other filters (Geneva et al, 2019a,b; Brossard et al, 2018). Geneva et al (2019a,b) leverage Schmidt Kalman formulation (SKF) (Schmidt, 1966) to treat old keyframes as nuisance parameters and thus reduce the quadratic computational complexity of EKF to $O(n)$. Brossard et al (2018) exploit the better nonlinear performance of UKF and embed states and errors into a high-dimensional Lie group. Others combine the optimization-based manner with MSCKF for comprehensive use of information. LOMSCKF (Heo et al, 2019) adopts optimization in the local window, which contains messages omitted in the original MSCKF such as camera observations. For a thorough understanding of MSCKF and its variants, OpenVINS (Geneva et al, 2020) provides an excellent open platform[11], which contains necessary resources ranging from documentation, tools, theories, to projects.

### 3.2.2 Optimization-based

The most dominant open-source optimization-based CI-SLAM systems currently are VINS-MONO (Qin et al, 2018) and its variant VINS-FUSION (Qin et al, 2019), ORB-SLAM3 (Campos et al, 2021), and DM-VIO (Stumberg and Cremers, 2022). The VINS (Qin et al, 2018, 2019) series are lightweight and easy to equip with embedded devices like Unnamed Aerial Vehicles (UAV). ORB-SLAM3 (Campos et al, 2021) integrates achievements of the ORB series (Mur-Artal et al, 2015; Mur-Artal and Tardós, 2017) and supports many modes such as monocular, stereo, their combination with IMU, and RGB-D. However, it relies more on visual features and is prone to fail in challenging environments (Yin et al, 2021). DM-VIO (Stumberg and Cremers, 2022) is based on DSO (Engel et al, 2018) and requires photometric calibration like all direct V-SLAM methods.

Commonly used sliding window-based BA (fixed-lag smoother) in optimization-based CI-SLAM linearizes the system by fixing linearization points. This leads to discrepancies with the original optimization and divergence. Besides, its marginalized priors are densely connected with remaining variables, imposing additional computing burdens, which can be amended by selective marginalization (Leutenegger et al, 2015; Qin et al, 2018) and sparsification of dense priors (Hsiung et al, 2018). In addition to improving the sliding-window, several methods increase the computational speed of BA. Liu et al (2018) offer an incremental BA-based solver and achieves state-of-the-art efficiency at that time. Gopinath et al (2023) establish GPU-accelerated BA and integrated it with ORB-SLAM3 with local-BA time reduction of 13.81%-33.79%.

[11] https://github.com/rpng/open_vins

### 3.2.3 Learning-based

Improvements in learning-based CI-SLAM typically exist in the feature extraction and message fusion stage. Some methods select or design feature extraction networks according to the features they prefer. Since CNN is not capable of capturing geometric features critical in state estimation, many learning-based CI-SLAM methods (Cai et al, 2020) have explored networks focusing on geometric features that can promote localization performance. Chen et al (2019) use FlowNet (Dosovitskiy et al, 2015) for geometrically meaningful image features rather than those related to appearance or context. As for IMU, Liu et al (2021) tailor a one-dimensional inertial feature encoder for IMU data to avoid the use of time-consuming LSTM.

In terms of fusion, instead of directly concatenating diverse features as in VINet (Clark et al, 2017), current approaches aim at designing a flexible fusion network. Chen et al (2019) add stochastic masks of features in the fusion stage and obtain the weights that overcome environment dynamics and data corruption. Besides, they demonstrate that IMU is more robust to fast rotation while visual measurements are preferred in large translations. This is beneficial for determining data ratios in fusion. Liu et al (2021) exploit attention-guided feature fusion that gets weight from cross-domain attention block and achieves better accuracy than conventional VIN-MONO and ORB-SLAM without loop detection. To raise the generalization of the model, researchers explore the self-supervised or unsupervised learning manners (Shamwell et al, 2018b; Han et al, 2019). Han et al (2019) take information from the stereo camera as the self-supervision but the feasibility relies heavily on the stereo camera.

As for the general challenges of CI-SLAM, CI-SLAM does not contain depth like LI-SLAM, which makes mapping more challenging. In addition, it contains more explorations of enhancing the utilization of IMU and motion patterns.

**Dynamic:** Different from DNN filters in LI-SLAM that simply removes potentially dynamic points, CI-SLAM further considers actual dynamics of detected objects (Song et al, 2022a; Samadzadeh and Nickabadi, 2023). Song et al (2022a) adaptively assign weights to features according to reprojection error values to reduce the impact of dynamic outliers. Besides, it proposes a multi-hypotheses-based constraints grouping method to avoid loop detection errors caused by temporal static objects. Samadzadeh and Nickabadi (2023) introduce HRNet (Yuan et al, 2020) as the feature evaluator to determine whether features are dynamic or static. In addition, using IMU measurements is less computationally intensive than using DNNs. Yu et al (2023) exploit IMU measurements in predicting the position of features and reducing outliers caused by dynamic environments. Zheng et al (2024) propose an IMU-based dynamic object tracker and combine object detection and bayesian filtering to find static points. Systems with stereo vision can leverage scene flow for dynamic detection (Yin et al, 2023a; Wen et al, 2024a). Yin et al (2023a) detect dynamic features by comparing IMU-prediction with stereo-generated scene flow. Then they construct virtual landmarks based on the detected dynamics, which are then used to improve the map construction process.

**Degeneracy** Similar to LI-SLAM, CI-SLAM systems also detect degeneracy through the quality of visual features (Yin et al, 2024). In addition to degeneracy caused by environments, researchers widely identify degenerative motion through observability analysis (Yang and Huang, 2018; Yang et al, 2019a). Since IMU measures the acceleration and angular velocity, it has no output in uniform linear motion and thus leads to divergence of the CI state estimator. Yang and Huang (2018) further derive that CI-SLAM will degenerate when the system moves along the direction of line features or rotates around the gravitational direction.

**Online Calibration** Several algorithms attenuate time offset through hardware and only obtain extrinsic online (Zou et al, 2019). Most methods optimize time offset and extrinsic at the same time by involving them in the state vector. Filter-based methods align time-offset-combined IMU propagation and visual pose estimation in the update stage (Eckenhoff et al, 2019a,b; Yang et al, 2019a). Eckenhoff et al (2019a) interpolate IMU pose into camera frames and estimate calibration parameters by representing the interpolation of unknown time offsets. Optimization-based methods establish factors relative to time and extrinsic (Qin and Shen, 2018). Qin and Shen (2018) propose visual feature factors composing time offset based on the constant velocity assumption.

**Feature:** Researchers abstract features for specific environments by leveraging their priors (Zou et al, 2019; Qin et al, 2020b; Chen et al, 2023a). For structural places, line features can be attached to the Atlanta world or calculated by vanishing points (Zou et al, 2019). Chen et al (2023a) propose a planar detection method special for man-made environments. Qin et al (2020b) deliberately detect semantic features that typically exist in parking lots such as guide signs, parking lines, and speed bumps for autonomous parking applications. IMU measurements are useful for narrowing down feature extraction and matching (Seok and Lim, 2020; Liu et al, 2023). Seok and Lim (2020) use inertial messages to generate soft-pose constraints in the initialization of feature tracking. Liu et al (2023) combine IMU in optical flow to predict the position of lines.

Not only features specific to particular environments, but visual cues with better quality also matter. Several methods ensure the quality by enhancing the tracking mechanism (Xie et al, 2020) or analyzing motion influences on features (Yu et al, 2023). Xie et al (2020) propose to seize the effective hierarchical quadtree search mechanism for robust feature tracking. Yu et al (2023) derive image perspective deformation models to construct an IMU-aided descriptor, which is robust to camera rotation. It is also effective to introduce a feature evaluation module (Zhang et al, 2023c; Jiang et al, 2023). Zhang et al (2023c) exploit recurrent field transforms from Teed and Deng (2021) to predict dense pixel-wise correspondences between co-visible views and attach confidences to each pixel. Jiang et al (2023) extract planer features from depth images and corner features from grayscale images. It contains a feature balancing module to identify whether it is a low-texture scene for feature choosing.

**Mapping:** The mapping thread of CI-SLAM systems commonly suffers from the sparsity of features and depth, which drives depth completion a key topic (Wong et al, 2020; Sartipi et al, 2020; Merrill et al, 2021). Deep learning methods are popular in depth completion (Wong et al, 2020; Sartipi et al, 2020; Merrill et al, 2021). Wong et al (2020) build a two-stage unsupervised dense depth completion network that combines piecewise planar and photometric consistency. Sartipi et al (2020) complete the depth of the low-texture surfaces from the sparse input and gravity direction. However, it can not meet real-time demand and only focuses on indoor planar environments. Merrill et al (2021) leverage Fastdepth (Wofk et al, 2019) and achieve real-time state estimation and depth completion. Some researchers exploit information obtained from IMU (Zuo et al, 2021) or the estimated odometry (Usenko et al, 2020). Zuo et al (2021) combined IMU measurements to provide metric dense depth reconstruction. Usenko et al (2020) recover non-linear factors from odometry and use them in the global consistent mapping optimization.

**Enhance utilization of IMU and motion patterns:** Compared to widely studied visual patterns, several recent approaches also take the IMU into account. Buchanan et al (2023) propose a learning-based IMU estimator that targets IMU bias instead of motion model and is, therefore, able to generalize to unseen locomotion patterns. SR-VIO (Samadzadeh and Nickabadi, 2023) adopts a denoising convolutional neural network to eliminate IMU noises and mitigate the divergence issue. The kinematic characters of the system offer additional constraints (Yu et al, 2021; Li and Stueckler, 2022). Yu et al (2021) combine constraints from motion information to confirm stop detection but is only suitable for ground vehicles. Li and Stueckler (2022) include velocity-control-based kinematic motion model (Thrun, 2002) constraints and further learns the parameters of it to predict the movement.

## 3.3 LiDAR-Camera SLAM

Since most LC-SLAM systems are designed in a loosely coupled manner and it is inconvenient to categorize them by fusion framework, we divided them into *Visual-centric* and *Modular-type* according to LiDAR utilization.

### 3.3.1 Visual-centric

Visual-centric LC-SLAM systems normally treat LiDAR merely as the range-augment instrument (Huang et al, 2020; Zhang et al, 2017; Graeter et al, 2018; Shin et al, 2018; Young et al, 2020; Zhu et al, 2021; Li et al, 2021a; Song et al, 2022b; Aydemir et al, 2022). Some methods use dense point clouds to improve depth registration accuracy (Young et al, 2020). DVL-SLAM (Shin et al, 2018; Young et al, 2020) is based on direct V-SLAM to relieve the depth association errors of 3D and 2D points in feature-based one. Extracting key features is also an effective approach to improving LC-SLAM. Huang et al (2020) introduce point and line features to ameliorate errors caused by noise, large viewpoint changes, and motion blurs.

### 3.3.2 Modular-type

Modular-type LC-SLAM systems not only combine the distance information but also resort to incorporating the function of LiDAR in SLAM systems by disassembling the systems into modular parts (Zhang and Singh, 2015; Seo and Chou, 2019; Wang et al, 2021c; Chen et al, 2021; Chou and Chou, 2022; Yin et al, 2022; Yuan et al, 2023; Shu and Luo, 2023; Cai et al, 2023). The notable work VLOAM (Zhang and Singh, 2015) is composed of a high-frequency LiDAR-assisted depth-enhanced VO and a low-frequency LO, refining the pose from the visual module. Later works tend to adopt the semi-direct V-SLAM methods instead of feature-based ones. SDV-LOAM (Yuan et al,

2023) and LAMV-SLAM (Yin et al, 2022) both based on semi-direct depth-enhanced odometry and pick intermediate frames to achieve better association between 3D and 2D points.

In modular-type LC-SLAM, cameras help LiDAR in many perspectives, such as color completion (Yin et al, 2022; Lin et al, 2024), loop closure (Wang et al, 2021c; Chen et al, 2021), mapping (Shu and Luo, 2023), and assitant constaint (Yin et al, 2022). Lin et al (2024) colored LiDAR point clouds by projecting RGB pixels to the LiDAR frame. Besides, they maintain a color feature map to constrain pose estimation along with the geometric map. Chen et al (2021) detect loop closure by the bag of world model in visual before point clouds verification, which compensates for LiDAR weakness in loop closure detection. Shu and Luo (2023) tailor a multi-modal mapping module that contains LiDAR and camera feature voxels. Yin et al (2022) comprise an online photometric calibration thread to eliminate photometric disturbances. Consolidated reference to the above systems, the general structure of the modular-type LC-SLAM can be illustrated in Fig.3.

In terms of common challenges, compared to the previous two types, LC-SLAM is concerned more with the data association process that connects LiDAR and camera measurements.

**Dynamic:** The dynamic objects can be rejected by alternatively adopting different types of sensors. TVL-SLAM (Chou and Chou, 2022) applies a multi-step outlier rejection mechanism. It uses LO as the initial pose to re-project visual map points onto the current image and then rejects matches with large re-projection errors.

**Online Calibration** Online calibration of LiDAR and the camera mainly focuses on extrinsic estimation. Some methods achieve this by matching heterogeneous features (Zhu et al, 2021). Zhu et al (2021) extract the depth, reflectivity, and contour of the LiDAR frame and then compare them with the image contour by ICP[12]. However, the feature-based calibration requires overlap in FOV, which is not always available. Thus, researchers also leverage the residuals of trajectory to optimize extrinsic parameters (Chou and Chou, 2022). For time offset, most LC-SLAM systems achieve synchronization by sophisticated hardware. Tu et al (2023) mitigate the need for absolute synchronization by splitting pose estimation into two steps. They initiate camera pose by global motion averaging and further joint optimize the separate poses.

**Data Association:** The data association is commonly implemented on the image plane (Graeter et al, 2018) or an intermediate frame, which is normally a unit sphere centered at the camera (Zhang et al, 2017; Yuan et al, 2023; Yin et al, 2022). LIMO (Graeter et al, 2018) first projects LiDAR points into the image plane to find a rough connections between visual and a set of point clouds. It then performs foreground segmentation to fit local planes and uses it to estimate accurate depth. In DEMO (Zhang et al, 2017), the extracted 2D features are associated with LiDAR depth by projecting them to a unit sphere for data association. Distinct from traditional handcrafted depth-association methods, learning-based depth completion is also studied (Aydemir et al, 2022; An et al, 2022; Liu et al, 2024). Aydemir et al (2022) propose a self-supervised hybrid fusion framework to decrease the rotation-only errors. An et al (2022) convert the point clouds to depth images and then feed them and RGB images

[12]https://github.com/ISEE-Technology/CamVox

into a pose estimation neural network. Liu et al (2024) propose a network with a bi-directional structure to fuse inconsistent modalities.

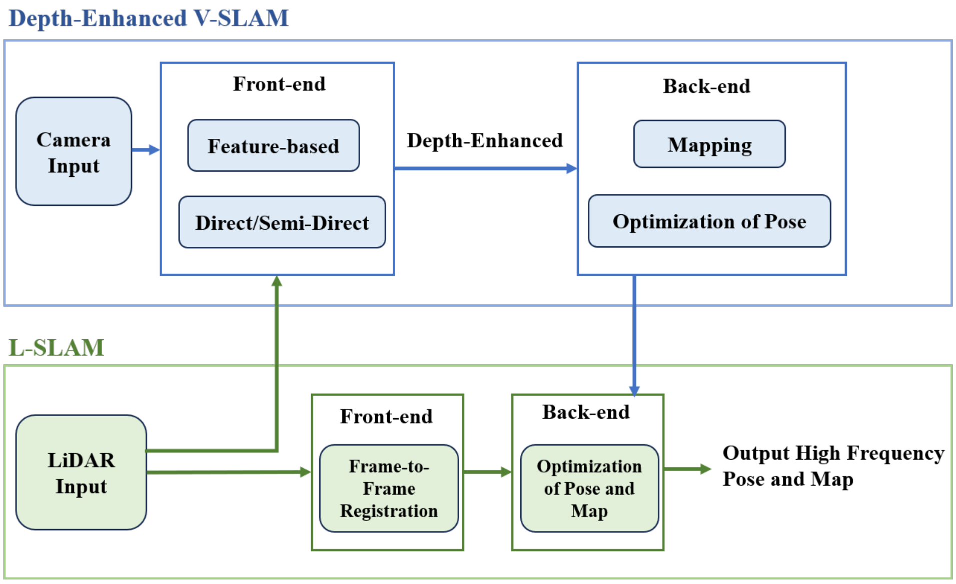


**Fig. 3** The general structure of Modular-type LC-SLAM systems.

### 3.4 LiDAR-IMU-Camera SLAM

LIC-SLAM systems are the integration of the aforementioned types and normally contain at least one subsystem of CI-SLAM or LI-SLAM. Early LIC-SLAM systems are mainly dominated by the loosely coupled manner and lack compact connections among diverse modalities (Wang et al, 2019; Khattak et al, 2020; Zhao et al, 2021; Khedekar et al, 2022). Several approaches reinforce the LI-SLAM with the help of CI-SLAM by initializing LiDAR mapping and scan matching using the estimated states of CI-SLAM (Wang et al, 2019; Khattak et al, 2020). VIL-SLAM (Shao et al, 2019) adopts CI-SLAM to initialize LiDAR mapping and leverages LiDAR-based pose estimation to enhance loop closure. It adopts the loosely coupled manner without any cooperation among measurements. More direct approaches parameterize the outputs of CI-SLAM and LI-SLAM as factors under the optimization-based fusion framework (Zhao et al, 2021; Khedekar et al, 2022). SuperOdometry (Zhao et al, 2021) proposes an IMU-centric framework that contains three sub-factor graphs for LI-SLAM, CI-SLAM, and LIC-SLAM. The IMU preintegration factor is used as motion prediction for LI-SLAM and CI-SLAM, and these two subsystems offer constraint factors to bind IMU biases.

Distinct from traditional fusion frameworks, Zhang and Singh (2018) propose a sequential pipeline that processes data in a decreasing frequency mode and formulates joint optimization as multiple small optimizations. The solution exploits a coarse-to-fine manner. The coarse estimation starts with IMU prediction and is then further refined by CI-SLAM and LI-SLAM. Despite the exclusivity of this pipeline, the subsequent tightly-coupled approaches are primarily grounded in filter-based or optimization-based implementations.

### 3.4.1 Filter-based

Filter-based methods remain similar to the previous process that propagates by IMU and updates with the LiDAR and camera measurements (Yang et al, 2019b; Zuo et al, 2019, 2020; Lin et al, 2021; Lin and Zhang, 2022; Zheng et al, 2022; Frosi and Matteucci, 2023). Zuo et al (2019) exploit the framework of MSCKF by preserving cloned past poses of camera and LiDAR. R2LIVE (Lin et al, 2021), R3LIVE (Lin and Zhang, 2022), and FAST-LIVO (Zheng et al, 2022) executes IESKF in the update stage. There is no explicit association between LiDAR and camera raw measurements in early LC-SLAM systems. The depth of visual measurements still originates from triangulation in (Zuo et al, 2019, 2020; Lin et al, 2021). Recently, R3LIVE and FAST-LIVO both establish connections between 3D LiDAR map and 2D visual images through photometric error and build a colored point cloud map of the environment. The photometric errors of R3LIVE (Lin and Zhang, 2022) are calculated by the residuals of RGB colors. FAST-LIVO (Zheng et al, 2022) is supported by FAST-LIO2 (Xu et al, 2022a) and SVO (Forster et al, 2014) with LiDAR map reuse. It avoids the triangulation and optimization of visual features over multiple keyframes. The specificities of ikdtree structure and the semi-direct method enable it to operate at high speed.

### 3.4.2 Optimization-based

Optimization-based approaches manage measurements of diverse stages as factors and achieve factor graph optimization (Wisth et al, 2021; Wang et al, 2021b; Jia et al, 2021; Shan et al, 2021; Tang et al, 2023; Wang and Ma, 2023; Wisth et al, 2023; Nam and Gon-Woo, 2023; Lang et al, 2023). The factor graph can be specially customized (Wang and Ma, 2023) or leverage canonical structures (Shan et al, 2021). mVIL-Fusion (Wang and Ma, 2023) proposes a three-level framework that contains a double-sliding-window optimization method. It assigns CI-SLAM as the main window and LiDAR data as the assist window with another window charging for loop closure and global mapping. LVI-SAM (Shan et al, 2021) is the combination of VINS-mono (Qin et al, 2018) and LIO-SAM (Shan et al, 2020) with visual depth augmentation by LiDAR. It improves the robustness of the initialization of VINS-MONO using calculations from LIO-SAM and involves a complete structure of degeneration detection, loop closure, and data association. Some methods adopt the continuous-time model that enables simultaneous motion distortion removal and trajectory estimation (Lv et al, 2023; Lang et al, 2023). CLIC (Lv et al, 2023) is the first method to utilize continuous-time fixed-lag smoothing with probabilistic marginalization in LIC-SLAM systems. Lang et al (2023) present non-uniform B-splines where control points are dynamically placed.

As for general challenges, LIC-SLAM incorporates the advantages of the former types and needs to consider the problems contained in each type.

**Degeneracy:** Analogous to solutions adopted in LI and CI SLAM, eigenvalues criterion (Shan et al, 2021; Khattak et al, 2020; Lee et al, 2024) and feature quality evaluation (Shan et al, 2021; Wang et al, 2021b) also work in the degeneracy detection of LIC-SLAM. For instance, CompSLAM (Khattak et al, 2020) detects the degenerate direction through eigenvalues of the Hessian of optimization function and changes to the CI submodule when LI degenerates. GR-Fusion (Wang et al, 2021b) exploits the intuitive method that rejects the LiDAR factor when the depth difference between the first and the last frame in the sliding window is less than the threshold. As for visual features, it considers the measurements unstable and ignores the visual factor when tracked features are lower than the threshold. Switch-SLAM (Lee et al, 2024) proposes a non-heuristic method to find the degenerate threshold of the LI submodule by Chi-squared test (Pearson, 1992). In addition, researchers migrate knowledge from other domains such as deep learning (Jia et al, 2021; Ju et al, 2022) and fuzzy logic (Nam and Gon-Woo, 2023). Livo (Jia et al, 2021) uses the actor-critic method in reinforcement learning to tune sensors' weight adaptively. Ju et al (2022) propose a learning-based scene-aware error model to evaluate the perception quality and use the result to promote the odometer performance. Nam and Gon-Woo (2023) exploit Type-2 fuzzy logic to build a learnable observation model that handles observation uncertainties and sensor failure issues.

**Feature:** Researchers consider commonalities between the same feature points captured by LiDAR and camera (Wisth et al, 2021, 2023). Wisth et al (2021) develop a unified multi-modal landmark tracking method for LiDAR and camera measurements, which ensures the uniform representation of 3D line and plane features. In VILENS (Wisth et al, 2023), they extend this representation and integrate it with leg odometry. Zuo et al (2020) propose a novel planar feature tracking method and divide planar features into SLAM landmark and MSCKF features. They treat those two types with Mahalanobis distance-based outlier rejection in LIC-Fusion2.0.

**Data Association:** Analogous to LC-SLAM systems, data association that assigns LiDAR depth to corresponding pixels matters in LIC-SLAM systems. The association patterns are also represented by projecting LiDAR point clouds into image planes (Wang et al, 2021b; Wisth et al, 2021) and projecting both measurements in a C-frame (Zhang et al, 2017; Chiodini et al, 2020; Zhang and Singh, 2018; Shan et al, 2021; Wang and Ma, 2023). A novel voxel-map-based depth-association method was proposed in a vanishing point-aided LIC-SLAM (Wang et al, 2021a). Besides, this work is further integrated into SuperOdometry (Zhao et al, 2021). CoCoLIC (Lang et al, 2023) obtains visual pixel depth from the global LiDAR map by formulating frame-to-map reprojection errors. Hence, it avoids the computationally intensive pixel depth estimation and optimization.

Except for correlating the depth information, handling occlusion or LiDAR stacking is also important in data association. Several methods reject outliers by checking the maximum distance among depth points for a feature (Zhang and Singh, 2018; Shan et al, 2021). FAST-LIVO (Zheng et al, 2022) first projects all the points in the visual submap and most recent LiDAR scan to the current frame. Then it examines their depth to check whether there is occlusion. In LE-VINS, Tang et al (2023) propose a solid state LiDAR-enhanced CI-SLAM which contains a novel depth association method by IMU. They notice that once the depth is obtained, it will be treated as a constant without any refinement or merely a prior in former researches. Hence they avoid outliers through an effective plane-checking algorithm.

In addition, there emerge some efforts to further enhance the correlation between data by exploiting visual measurements (Lin and Zhang, 2022; Zheng et al, 2022; Hong et al, 2024). The vectors of tracked points in R3LIVE (Lin and Zhang, 2022) contain their 3D position and RGB color. The frame-to-map photometric update relies on the residuals between the predicted and the observed color. After the system converges, the optimized color will be adopted in rendering the texture of the global map. FAST-LIVO (Zheng et al, 2022) attaches patches from the current image to map points within the FOV. LVI-GaussMap (Hong et al, 2024) achieves high-fidelity map reconstruction. It refines the Gussian LiDAR map by visual photometric gradients.

**Online Calibration:** Several LIC systems take IMU as the mediator and obtain calibration of IMU with LiDAR and camera respectively (Zuo et al, 2019, 2020). Zuo et al (2019, 2020) propagate IMU measurements up to the corrected time and involve the extrinsic in the measurement models of LiDAR and camera. Some LIC systems divide calibration into two phases. Yang and Ma Wang and Ma (2022) first calibrate the CI submodule and then linearly interpolate CI-based pose estimation onto the LiDAR for LI calibration. R3LIVE (Lin and Zhang, 2022) only calibrates IMU and camera online by containing time offset in the measurement model of the camera. Tong et al (2024) propose a novel keyframe-oriented data synchronization method to align LiDAR and camera measurements.

## 3.5 Summary

The summarization of the four types of multi-sensor fusion SLAM is shown in Table 1. Through the above analysis, it can be concluded that in addition to refinements of fusion frameworks, multi-sensor fusion SLAM systems commonly concentrate on dynamic processing, degeneracy treatment, online calibration, and feature extraction. CI-SLAM involves more settlements for mapping and enhancing the utilization of IMU. LC-SLAM and LIC-SLAM additionally concern data association between LiDAR and camera measurements.

### 3.5.1 Dynamic Detection and Processing

Environment dynamics mainly influence observations of LiDAR and cameras. Both of them can find dynamic points by DNN (Zhao et al, 2019; Samadzadeh and Nickabadi, 2023) or geometric constraints (lio, 2021). Containing multi-sensors, dynamic detection can also be achieved by comparing differences of individual measurements (Yin et al, 2023a). The treatment of dynamic has improved in recent years, which ranges from directly filtering out dynamic points (Zhao et al, 2019), exchanging diverse modalities (Seo and Chou, 2019; Samadzadeh and Nickabadi, 2023), considering temporarily static points (Song et al, 2022a), to maintaining dynamic points as a constraint (Yin et al, 2023a).

### 3.5.2 Degeneracy Detection and Processing

Degeneracy can be caused by environments and motions. General fusion frameworks can alleviate it by covariance matrix of the measurements to some extent. However, this approach will fail when the degeneration is too severe to lead to system breakdown.

Environmental degeneracy influence the performance of LiDAR and camera and is normally detected through eigenvalues of the Hessian matrix of the measurements equation as proposed by Zhang et al. (Zhang et al, 2016). The degenerate directions are related to the eigenvectors corresponding to eigenvalues smaller than the threshold, which is defined through experiments (Khattak et al, 2020; Hinduja et al, 2019; Tagliabue et al, 2021), heuristic methods (Han et al, 2023), and non-heuristic method (Lee et al, 2024). The more intuitive way is checking the quality and number of features (Wang et al, 2021b; Shan et al, 2021; Júnior et al, 2022). Some systems also consider message rate (Palieri et al, 2021). In addition, multi-sensor systems can rely on measurements from other sensors. For instance, LIVER leverages CI-SLAM to determine the degenerate directions of LiDAR, while Ground-Fusion (Yin et al, 2024) adopts the wheel odometer for visual degeneracy detection. Modern systems also introduce DNN-based degeneracy detection modules (Jia et al, 2021; Ju et al, 2022; Nam and Gon-Woo, 2023). Motion-caused degeneracy has been widely researched in aided-IMU systems by analyzing the null space of the observability matrix (Yang and Huang, 2018). As for treatments of degeneracy, some systems simply invalidate data from sensors that are detected to degenerate (Palieri et al, 2021; Tagliabue et al, 2021; Shan et al, 2021; Yin et al, 2024) and constraint the degenerative motions (Yang and Huang, 2018; Yang et al, 2022). Systems that detect specific degeneracy directions may optimize the state along with the well-constrained directions (Wen et al, 2024b).

### 3.5.3 Feature Extraction

The extraction of features depends on sensing modalities and the environment prior. There are numerous mature approaches in computer vision, such as corner features (Shi et al, 1994; Leutenegger et al, 2011), edge features (Canny, 1986), and line feature detector (Grompone von Gioi et al, 2010). Although researchers have contributed many novel definitions and descriptions of point, line, and plane features, exploring potential features that are relevant for state estimation is still meaningful. Features existing in a particular scenario can also enhance the performance of the system, such as structural feature extraction in artificial scenes.

### 3.5.4 Online Calibration

Online calibration generally adds extrinsic and time offset parameters into the state vector to optimize together with the pose of the system. For IMU-included systems, original calibration methods minimize the pose estimation discrepancy between IMU propagation and LiDAR or camera odometry (Tagliabue et al, 2021; Le Gentil et al, 2021; Qin and Shen, 2018; Eckenhoff et al, 2019a,b; Yang et al, 2019a; Zou et al, 2019; Zuo et al, 2019, 2020; Wang and Ma, 2022; Lin and Zhang, 2022). Several methods are refined to residuals of angular velocity and acceleration (Zhu et al, 2022a). Most sensor drivers nowadays guarantee a rough clock source synchronization under network protocols once it is plugged into the system. Online calibration allows for more accurate time synchronization. The time offset is always included in measurement functions, such as IMU propagation (Le Gentil et al, 2021) and camera measurement (Qin and Shen, 2018; Lin and Zhang, 2022) to ensure the time of different sensors is consistent. For LC-SLAM, online calibration mainly focuses on extrinsic estimation through the matching between LiDAR and camera observations (Zhu et al, 2021).

### 3.5.5 Heterogeneous Data Association

Data association is mainly the problem in LC-SLAM and LIC-SLAM systems that require the correlation of LiDAR and camera data. Ranging from the simple projection of LiDAR point cloud to image plane (Graeter et al, 2018), optimization based on residuals between 3D map and 2D pixels (Lang et al, 2023), to correlation involving RGB information (Lin and Zhang, 2022). The interaction is becoming tighter and more robust.

**Table 1**: The comparison of different types of algorithms.

| Type | Reference | Framework | Contribution | Code | Hardware |
|---|---|---|---|---|---|
| LI-SLAM | LIO-SAM (2020) | factor | local sliding window-based scan-matching to enable real-time performance | ✓ | VLP16, MicroStrain 3DM-GX5-25 IMU; Intel i7-10710U CPU |
| | CLINS (2021) | factor | non-rigid registration for continuous-time trajectory; two-stage trajectory correction | ✓ | VLP-16 LiDAR, Xsens-300 IMU |
| | FAST-LIO2 (2022a) | IESKF | directly register scan and map; ikd-tree for map maintenance | ✓ | Livox Avia, LiDAR's built-in IMU; DJI Manifold-2 C |
| | Faster-LIO (2022) | IESKF | present iVox for point clouds; AA-IESKF; iVox-PHC for bunch of points | ✓ | - |
| | Point-LIO (2023) | EKF | high bandwidth; stochatic process-augmented kinematic model | ✓ | Livox Avia, LiDAR's built-in IMU; Khadas VIM3 Pro |
| | D-LIOM (2023b) | factor | combine gravity as factors; submap-to-submap loop detection | ✓ | 16line ROBOSENSE, ZED's built-in IMU; Intel Xeon E5-2620V3 CPU |
| | TransFusionOdom (2023) | learning | leverage advantages of Transformer | ✗ | - |
| | LIO-GVM (2024) | IESKF | incremental Gaussian voxel map; variance residual | ✓ | - |

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| Type | Reference | Framework | Contribution | Code | Hardware |
|---|---|---|---|---|---|
| CI-SLAM | VINS-MONO (2018) | factor | representative framework covering sliding-window based initialization, optimization and loop detection | ✓ | MatrixVision mvBlueFOX-MLC200w, DJI A3 built-in IMU; Intel i7-5500U CPU |
| | DeepVIO (2019) | leaning | self-supervised learning-based method | ✗ | - |
| | ORB-SLAM3 (2021) | factor | fast IMU initialization; novel place recognition algorithm; multimap system; abstract camera representation | ✓ | - |
| | AT-VIO (2021) | learning | one-dimension IMU encoder; attention block for fusion | ✗ | - |
| | DM-VIO (2022) | factor | delayed marginalization and PGBA for IMU initialization; dynamic photometric error weight | ✓ | - |
| | Eq-VIO (2023) | EqF | EqF; novel symmetry for VI-SLAM; reduce linearization error | ✓ | - |
| | SR-VIO (2023) | factor | hybrid visual preprocessing; switchable factors; transformer-based loop clousure | ✗ | - |
| | RLD-SLAM (2024) | factor | IMU-based dynamic object tracker; object detection and bayesian filtering for static points | ✗ | ZED2 stereo camera, control board's IMU; Intel i7 CPU, NVIDIA GeForce GTX 1050 Ti GPU |

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| Type | Reference | Framework | Contribution | Code | Hardware |
|---|---|---|---|---|---|
| LC-SLAM | CamVox (2021) | - | LiDAR and camera extrinsic calibration; solid LiDAR | ✓ | Livox Horizon, MV-CE060-10UC camera; DJI Manifold 2C |
| | PanoVLM (2023) | - | panoramic vision and LiDAR fusion no need for strict synchronization | ✓ | VLP-16, Insta360 ONE X2; Raspberry Pi 4B |
| | SDV-LOAM (2023) | - | new feature extraction and point matching method; semi-direct depth enhance | ✓ | VLP-16, FL3-U313E4M-C; Intel Core i7-11700 |
| | Lin et al (2024) | - | RGB colored dense point clouds color and geometric feature constrains | ✗ | solid-state LiDAR, Intel RealSense L515; AMD Ryzen 9 7900X 12-Core CPU |
| | DVLO (2024) | - | end-to-end LC-SLAM local-to-global fusion network | ✗ | - |
| LIC-SLAM | LIC-Fusion2.0 (2020) | EKF | novel plane-feature tracking; observability analysis of LIV; outlier rejection | ✗ | VPL-16, Xsens IMU, monocular camera; Intel i78086k CPU |
| | LVI-SAM (2021) | factor | failure detection; loop closure; well opensourced structure | ✓ | VLP-16, MicroStrain 3DM-GX5-25 IMU, FLIR BFS-U3-04S2M-CS camera; Intel i7-10710U |

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| Type | Reference | Framework | Contribution | Code | Hardware |
|---|---|---|---|---|---|
| | R3LIVE (2022) | IESKF | LIO for map's geometric structure; VIO render map's texture; 3D map-2D image error; online calibration, RGB contained | ✓ | LiVOX AVIA, LiDAR built-in IMU, FLIR Blacky BFS-u3-13y3c camera; DJI Manifold 2C |
| | mVIL-Fusion (2023) | factor | VIO-based LO prediction; double sliding-window; software time sync | ✓ | LeiShen-C16 spinning LiDAR, MYNTEYE-D120 VI sensor; DJI Manifold 2C |
| | Dinh and Kim (2023) | factor | AI-assited noise model; Type-2 Fuzzy logic in factor | ✗ | Hokuyo UTM-30LX LiDAR, MTi-100IMU, Zed 2 camera; NVIDIA Jetson AGX Xavier |
| | Coco-LIC (2023) | factor | non-uniform B-spline; image-to-lidar reprojection | ✓ | VLP-16, Xsens MTi-300 IMU; Intel i7-8700 CPU |
| | LIV-GaussMap (2024) | IESKF | high fidelity Gaussian map; compatible with various types of LiDAR | ✓ | LiVOX AVIA, LiDAR built-in IMU, MV-CA013-21UC; Intel Core i9 12900K, NVIDIA GeForce RTX 4090 |
| | Switch-SLAM (2024) | factor | non-heuristic degeneracy detection, switching-based sensor fusion | ✗ | Intel i7-1165G7 CPU |

**Table 2** Datasets for multi-sensor fusion SLAM

| Datasets | Sensor | | | | Environment | | Collected | Dynamics | Degeneracy |
|---|---|---|---|---|---|---|---|---|---|
| | LiDAR | IMU | Camera | Other | Indoor | Outdoor | | | |
| KITTI(2012) | HDL-64E | ✓ | ✓ | GNSS | - | Urban | Car | medium | highway; high speed |
| Oxford Robotcar(2017) | HDL-64E | ✓ | ✓ | - | - | Urban | Car | medium | long-term weather changes; building variations |
| NCLT(2016) | HDL-32E; UTM-30LX | ✓ | ✓ | GNSS | Campus | Campus | GR | high | changing light levels; snow |
| EuROC(2016) | - | ✓ | ✓ | - | Machine Hall; Vicon room | - | UAV | - | sharp turns; bobbing up and down |
| TUM-VI(2018) | - | ✓ | ✓ | MoCap; Light sensor | Campus | Campus | Hand | - | corridors, in-out changes |
| Argoverse(2019) | VLP-32C | ✓ | ✓ | GNSS | - | Urban | Car | high | textureless openspace |
| Nusecens(2020) | HDL-64E | ✓ | ✓ | GNSS; Radar | - | Urban | Car | high | rainy, snow, high speed |
| 4Seasons(2021) | - | ✓ | ✓ | GNSS | Parking lot | Urban; Highway; Rural | Car | high | challenging weathers; tunnels and highways |
| Urbanloco(2020) | HDL-32E | ✓ | ✓ | GNSS | - | Urban | Car | medium | very tall buildings; urban canyons, bridges, and tunnels; driving on cross-harbour road |
| Newer College(2020) | Ouster-64 | ✓ | ✓ | - | - | Campus | Hand | - | textureless open space; aggressive angular motions |
| UtbmRobotcar(2020) | HDL-32E | ✓ | ✓ | GNSS; Radar | - | Urban | Car | high | snow, sloping road, right overtaking |
| VIODE(2021) | - | ✓ | ✓ | - | Parking lot | City | UAV | high | extensive FOV obstruction |
| M2DGR(2021) | VLP-32 | ✓ | ✓ | Infrared camera; Event camera; GNSS | Campus | Campus | GR | low | back-and-forth;zig-zag; lift; textureless hall |
| NTU-VIRAL(2022) | Ouster-16 | ✓ | ✓ | UWB | Campus | Campus | UAV | - | 6DOF drastic movements; |
| Hilti-Oxford(2023b) | Hesai-32 | ✓ | ✓ | - | Buildings | - | Hand | low | dark corners, long corridors, narrow stairs |
| FusionPortableV2(2024) | Ouster-128 | ✓ | ✓ | Event camera | Campus | Campus | Hand; QR; Car | high | grassland, escalator, underground tunnel |
| MCD(2024) | OS1-64 Livox Mid-70 | ✓ | ✓ | UWB | - | Campus | Hand | low | glass reflection, extreme lighting |

# 4 Datasets, Performance Evaluation, and Hardware Configuration

## 4.1 Dataset

Open-source datasets avoid complex hardware preparation and deliver a uniform platform to compare the algorithms in diverse scenarios. We collate frequently used datasets in multi-sensor fusion SLAM in Table2.

Most datasets contain all of the three sensors (Geiger et al, 2012). Some datasets that have been widely used in CI-SLAM evaluation are collected only by cameras and IMU (Burri et al, 2016; Schubert et al, 2018; Wenzel et al, 2021; Minoda et al, 2021). The degeneracy conditions, such as degenerative scenarios and aggressive motion are generally included in current datasets.

Some datasets are oriented toward autonomous driving scenarios and collected by high-speed cars (Geiger et al, 2012; Maddern et al, 2017; Wen et al, 2020; Wenzel et al, 2021). KITTI (Geiger et al, 2012) is a classic dataset collected by a passenger vehicle, involving urban scenes and textureless highways. Oxford RobotCar dataset (Maddern et al, 2017) contains 1-year driving data, involving long-term weather changes and building variations. UrbanLoco (ulhk) (Wen et al, 2020) is collected in highly urbanization areas, containing urban canyons, bridges, and tunnels. 4Seasons (Wenzel et al, 2021) contains 350km of recordings in nine environments, ranging from tunnels to highways. Some datasets are collected by a relatively slow Ground Robot (GR) (Carlevaris-Bianco et al, 2016; Yin et al, 2021). North Campus Long-Term (NCLT) dataset (Carlevaris-Bianco et al, 2016) is collected on a campus covering 147.4 km

**Table 3** Performance Comparison of multi-sensor fusion SLAM Algorithms

| Type | Method | Exchange | Indoor | | | | | | Outdoor | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | door1 | room1 | room3 | lift1 | lift3 | hall2 | hall4 | gate1 | gate3 | street1 | street3 | street7 | walk1 |
| LI | Faster-LIO (2022) | 0.410 | 0.396 | 0.429 | 0.886 | 1.709 | **0.565** | 1.068 | 0.162 | 0.201 | **0.346** | 0.149 | **11.891** | **3.301** |
| CI | VINS-MONO (2018) | 1.811 | 0.342 | 0.472 | **0.422** | 5.017 | 1.227 | 2.058 | 3.688 | 7.555 | 67.036 | 5.394 | 122.040 | 9.725 |
| LIC | LVI-SAM (2021) | **0.265** | **0.153** | **0.161** | 1.349 | **0.797** | 1.814 | **0.953** | **0.145** | **0.109** | 0.753 | **0.134** | 12.163 | 3.370 |

over 15 months. M2DGR (Yin et al, 2021) contains challenging situations not included in previous datasets, such as entering lifts and complete darkness. Beyond that, it involves sequences with back-and-forth and zig-zag movements. Datasets collected via UAV (Burri et al, 2016; Nguyen et al, 2022) or handheld devices (Schubert et al, 2018; Ramezani et al, 2020; Wei et al, 2024) deliver more motion patterns. NTU-VIRAL (Nguyen et al, 2022) is obtained by UAV on campus, containing trajectories such as UAV movement as it turns and rises. Newer College (Ramezani et al, 2020) involves data of highly dynamic motions in textureless open space. FusionPortableV2 (Wei et al, 2024) contains diverse platforms ranging from a handheld device, a quadruped robot (QR), and a vehicle. Hilti-Oxford dataset (Zhang et al, 2023b) is collected by handheld device in indoor environments, containing floor variations and limited texture.

There are varying levels of dynamics that exist in datasets. Autopilot scenarios naturally include dynamic objects. However, most of them contain only medium levels of dynamic disturbances, with occasional pedestrians or vehicles passing by (Geiger et al, 2012; Maddern et al, 2017). Some datasets are less dynamic, with only some sequences containing slow-moving objects (Yin et al, 2021; Zhang et al, 2023b). VOIDE (Minoda et al, 2021) is captured in simulation and primarily used to evaluate methods incorporated with dynamic processing. UtbmRobocar dataset (Yan et al, 2020) contains cyclical variation and highly dynamic objects. In addition, some datasets contain semantic information that can support intelligent SLAM methods (Geiger et al, 2012; Maddern et al, 2017; Caesar et al, 2020; Chang et al, 2019). Nuscenes (Caesar et al, 2020) and Agroverse (Chang et al, 2019) offer information on maps, 3D bonding boxes, and trajectories in city driving scenarios. Recent datasets have started to involve measurements of solid-state LiDAR. For instance, Multi-Campus Dataset (MCD) (Nguyen et al, 2024) contains scans of Livox Mid-70.

## 4.2 Performance Evaluation

Different types of multi-sensor fusion SLAM systems are generally tested on different datasets, making it difficult to assess their performance from data in papers alone. Thus, to help readers build an intuitive understanding of the performance of these algorithms, we compare several traditional algorithms on the M2DGR dataset Yin et al (2021) and collate the performance of some learning-based algorithms on the KITTI dataset Geiger et al (2012).

**Table 4** Performance Comparison of multi-sensor fusion SLAM Algorithms

| Type | Method | 01 | | 05 | | 08 | | 09 | | 10 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $t_{rel}$ | $r_{rel}$ | $t_{rel}$ | $r_{rel}$ | $t_{rel}$ | $r_{rel}$ | $t_{rel}$ | $r_{rel}$ | $t_{rel}$ | $r_{rel}$ |
| LI | DeepLIO (2021) | 5.28 | 1.51 | 1.28 | 0.93 | 2.33 | 0.94 | 4.4 | 1.21 | 4.0 | 1.51 |
| | TransfusionOdom (2023) | **0.43** | 0.46 | 0.48 | 0.73 | **0.99** | 0.75 | **0.49** | 0.63 | **0.72** | 0.78 |
| CI | DeepVIO (2019) | - | - | 2.86 | 2.32 | 2.13 | 1.02 | 1.38 | 1.12 | 0.85 | 1.03 |
| | ATVIO (2021) | - | - | 1.97 | 0.98 | 2.04 | 0.88 | 3.99 | 1.63 | 4.06 | 1.81 |
| | EMA-VIO (2022) | - | - | - | - | - | - | 8.68 | 1.54 | 7.46 | 2.66 |
| LC | An et al (2022) | 2.53 | 0.79 | 3.49 | 0.79 | 2.75 | 1.61 | 3.70 | 1.83 | 5.65 | 0.51 |
| | H-VLO (2022) | 1.75 | 0.62 | 0.85 | 0.35 | 1.35 | **0.38** | 1.89 | 0.34 | 1.39 | 0.52 |
| | DVLO (2024) | 0.85 | **0.33** | **0.41** | **0.23** | 1.09 | 0.44 | 0.85 | **0.36** | 0.88 | **0.46** |

### 4.2.1 Evaluation Metrics

The evaluation metrics of a SLAM system focus on its accuracy and efficiency. The efficiency is evaluated by the runtime of the algorithm. For accuracy, the Absolute Trajectory Error (ATE) and the Relative Pose Error (RPE) are commonly used (Schubert et al, 2018). ATE measures the global deviation between the estimated trajectory and the true trajectory, which is useful to test the global consistency. RPE quantifies the local accuracy of the trajectory within a fixed time interval $\Delta$, such as revealing local drift per meter. They are both calculated by the root mean square error (RMSE) and can be shown as:

$$\begin{aligned}
ATE_{RMSE} &= \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left\|trans(T_{gt,i}^{-1} * T_{est,i})\right\|^2} \\
RPE_{RMSE} &= \sqrt{\frac{1}{N-\Delta}\sum_{i=1}^{N-\Delta}\left\|trans(T_{gt,i,\Delta}^{-1} * T_{est,i,\Delta})\right\|^2} \\
T_{gt,i,\Delta} &= T_{gt,i}^{-1} * T_{gt,i+\Delta} \\
T_{est,i,\Delta} &= T_{est,i}^{-1} * T_{est,i+\Delta}
\end{aligned} \tag{7}$$

where $T_{gt,i}$ and $T_{est,i}$ represent the real and estimated pose at i-th timestamp. Learning-based methods are normally tested on KITTI (Geiger et al, 2012) and use $t_{rel}(\%)$ (The average translational RMSE drift (%) on lengths of 100m-800m) and $r_{rel}(\%)$ (the average rotational RMSE drift (°/100m) on lengths of 100m-800m) as evaluation metrics. The mapping performance is still qualitatively judged by manual observation. With the emergence of methods with 3D Gaussian and Nerf, Peak Signal to Noise Ratio (PSNR) and Structural Similarity Index (SSIM) are also adopted in mapping evaluation (Hong et al, 2024).

### 4.2.2 Performance Comparison

The sequences we choose from M2DGR (Yin et al, 2021) contain indoor, outdoor, and in-to-out conditions, while KITTI (Geiger et al, 2012) used in learning-based methods only contains outdoor driving data.

**Conventional Algorithms:** We use ATE (m) for the comparison of the location accuracy of multi-sensor fusion SLAM algorithms by EVO[13] tool as shown in Table 3. Systems containing LiDAR perform better than the CI-SLAM method. Especially in outdoor environments, the accuracy of LI and LIC SLAM is not in the same class as that of CI-SLAM. This reflects that LiDAR can provide more reliable measurements than cameras in outdoor applications. *street1* and *street7* are long-term sequences with a zig-zag motion. The CI-SLAM method produces large drifts on both of them, which shows that the long-term stability of CI-SLAM still needs to be enhanced. In addition, all methods are challenged in *street7*. We believe that this is because the sequence contains many sharp turns. This suggests that cornering is a challenging moving pattern for SLAM systems. In indoor environments, these methods perform similarly, thus CI-SLAM is more suitable for indoor applications since it always contains the lowest hardware cost. In almost all sequences, LIC-SLAM achieves the best performance. On individual sequences where it did not behave best, its performances are also not far from the best method. This demonstrates the robustness and stability of the LIC-SLAM system.

**Learning-based Algorithms:** We use data from (Sun et al, 2023; Liu et al, 2021, 2024; Tu et al, 2022) to compare the performance of learning-based algorithms in the three frameworks, as shown in Table4. LI and LC methods can reach errors within 1 meter and 1°. CI methods are less favorable, reaching errors within 5m and 2°. LI method is better in translation, while the LC method is better in rotation. It is understandable since the combination of LiDAR and IMU brings better distance information, benefiting translation estimation.

## 4.3 Hardware Configuration

We summarize the hardware configuration and computational energy for algorithms tested with self-assembled hardware in Table1. The self-assembled devices are usually fixed to a metal frame and can be ported to different unnamed systems. The most chosen spinning LiDAR is from Velodyne and solid-state LiDAR is from Livox. Which type of LiDAR to pick only depends on the algorithm. Some researchers choose industrial IMU (Zuo et al, 2020; Lv et al, 2021), while others only use IMU built inside LiDAR or cameras (Bai et al, 2022; Lin and Zhang, 2022). The latter reduces cost and calibration consumption without much impact on the accuracy of the algorithm. Although those physical experiments do not reflect a general preference for camera models, most researchers use global shutter monocular ones. For computational requirements, most algorithms are deployed with an onboard PC, such as DJI Manifold 2-C[14], Raspberry Pi 4B[15], and NVIDIA Jetson AGX Xavier[16]. Eyvazpour

[13]https://github.com/MichaelGrupp/evo
[14]https://www.dji.com/cn/manifold-2
[15]https://www.raspberrypi.com/products/raspberry-pi-4-model-b/
[16]https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-agx-xavier/

et al (2023) review the hardware implementation of SLAM systems, especially focus on the computation devices.

# 5 Future Challenges

The following section proposes several future challenges to formulate a more versatile multi-sensor fusion SLAM system.

## 5.1 Make full use of deep neural networks

DNNs have the potential to extract the implicit connection between a diversity of data, which is revealed in other fields (Chitta et al, 2022). It inspires us to explore their possible capacity in multi-sensor fusion SLAM systems.

### 5.1.1 Combination of geometric and semantic feature

Traditional methods merely extract artificial features represented by points, lines, and planes. These features are not stable in unstructured or resolution-changing scenes (Lianos et al, 2018) and are deficient in environmental comprehension. However, DNNs can grasp both geometric and semantic features according to specialized loss functions. For geometric clues, DNNs generally capture the optical flow in the image stream (Teed and Deng, 2021; Zhang et al, 2023e), which still lacks the understanding of the scenarios. Semantic clues exploited in semantic SLAM systems can facilitate dynamic filtering and loop detection but are commonly parallel with the geometric thread, omitting its contribution to localization. Thus, simultaneously extracting geometric and semantic features by a network enables achieving an end-to-end multi-sensor fusion SLAM system with scene interpretation. Moreover, DNNs have been productive in multimodal data processing in other domains, such as object detection and semantic segmentation (Chitta et al, 2022; Zhang et al, 2023a). The output of these DNNs and the ideas they employ can be migrated into multi-sensor fusion SLAM. Nonetheless, the generalization of learning-based feature extraction is affected by datasets. The more datasets DNNs are trained on, the more scenarios they can adapt to. However, the actual computing power is limited. The Large Model[17] may address this issue to some extent.

### 5.1.2 Pose Regression

Pose regression means regressing the pose by DNNs, which is first practiced by PoseNet (Kendall et al, 2015). Researchers improve it by adding LSTM layers (Walch et al, 2017) or exploiting Transformer (Shavit et al, 2021) in pose inference. Sattler et al (2019) reveal that pose regression is more related to image retrieval and desires refinements for better accuracy. Data from various sensors may contribute to enriching the statistical support for pose regression and obtaining better results.

[17] https://chat.openai.com/

## 5.2 Enhance and exploit map information

The present research prefers strengthening the pose estimation and often neglects the enhancement of the mapping function with multi-sensors.

### 5.2.1 Map Quality

Traditional map representations are discrete and can be categorized as depth point, height, surfel, and volumetric. With the development of implicit mapping and neural radiance fields (NeRF) (Mildenhall et al, 2021), several V-SLAM algorithms employ this continuous representation to render the scenarios (Zhu et al, 2022b; Zhang et al, 2023e). GO-SLAM (Zhang et al, 2023e) enables real-time implicit neural mapping in large-scale indoor environments and achieves more prise localization results on some datasets where ORB-SLAM3 fails. Recent works also combine 3D Gaussian splatting (Hong et al, 2024).

### 5.2.2 Map Application

Since most systems solely leverage the maps in global optimization during a single run, the subsequent value of maps obtained by multi-sensor SLAM systems is seldom mentioned. These maps contain ample metric or even semantic details and can be branched out to multiple downstream missions, ranging from navigation and post-disaster exploration to augmented reality human-computer interaction. Besides, the multi-sensor configuration can support a more comprehensive perception of the world. Arm et al (2023) exploits the multi-sensor fusion SLAM algorithm in the scout robot which is in charge of exploring the planetary analog environment. The robot offers the environmental map for other robots on planetary exploration, demonstrating the promising scope of expanding applications of maps reconstructed via SLAM.

## 5.3 Fuse more modalities and analyze optimal combinations

Although the integration of LiDAR, IMU, and camera is the most common configuration in present multi-sensor fusion SLAM systems, combinations with other modalities can provide information unavailable in current systems. The extensible modalities are pretty diverse, ranging from GNSS (Cao et al, 2022), Ultra Wide Band (Nguyen et al, 2022), to event cameras (Zhou et al, 2021). Especially with the widespread implication of quadrupedal robots and wheeled vehicles, the combination with wheel encoders (Hua et al, 2023) and leg odometry (Yang et al, 2023) has become increasingly popular. Apart from that, invoking sensors from neighboring infrastructures also has the potential to promote multi-sensor fusion SLAM systems. For instance, in places supported by wireless communication signals, robots performing exploration tasks can access information from surveillance cameras in the buildings as an auxiliary to the SLAM system.

However, while it is de-facto that multi-sensor fusion SLAM systems deliver better performance than those equipped with fewer sensors, the noise and computational burden that accompany multiple modalities should not be overlooked. We believe that there exist optimal combinations of modalities, both in terms of the number and types of them. Existing theoretical analyses primarily concentrate on the consistency and invariance of the fusion framework or dependent features. They yield the optimal solution in a specific manner rather than investigating the necessity of a supplementary sensor. Hence, to prevent the abuse of sensors and maximize the effectiveness of each sensor, it is imperative to analyze whether there is an upper threshold to the benefits afforded by additional modalities and where the threshold lies.

## 5.4 Design novel general fusion framework

A critical subject in multi-sensor fusion SLAM systems is determining how to integrate a wide range of sensor measurements, namely the selection of a fusion framework. The conventional fusion frameworks based on filtering or BA have dominated and spawned a series of accurate and efficient algorithms with the learning-based fusion progressively coming to prominence. However, these types of fusion frameworks prioritize purely perceptual effects and neglect the requirements of the missions as well as the limitations of the scenarios. Hence, designing a task-driven fusion framework that adaptively configures and selects the sensors suitable for the demands of scenarios and assignments is essential.

## 5.5 Combination of another application

The extent application of SLAM can cooperate with object tracking that contains a mutual facilitating effect, which is called SLAMMOT (Feng et al, 2023; Lin et al, 2023). By denoting the dynamic obstacles in bounding boxes, the system leverages the dynamic elements to assist self-localization rather than directly discarding them. Meanwhile, the SLAM results will correspondingly boost the performance of object tracking, empowering the system to fulfill more intricate tasks, i.e. decision and planning in autonomous driving.

Henning et al (2023) further broadens this incorporation to binding with human motion tracking and achieves better performance than ORB-SLAM3 in datasets containing people. Not only vehicles and humans but virtually all kinetic objects in the environment can be exploited, and it is an open problem of developing a generic paradigm for modeling them and what trade-offs ought to be imposed for various dynamic objects. Air-ground coordination and cross-view matching (Shi et al, 2019) are also considerable extensions since they can offer extra localization and environmental information.

## 5.6 Reduction of computational consumption

SLAM algorithms are often supported as localization modules for unmanned systems. However, the systems may also contain modules for decision-making, batting, and planning. These modules share the arithmetic power of the on-board computer, leading to the problem of resource hogging. Thus it is necessary to design multi-source fusion SLAM algorithms with high computational efficiency and low energy consumption. It provides stable localization and map-building support for unmanned systems with minimal consumption.

# 6 Conclusion

This paper is intended to provide an overview of research in multi-sensor fusion SLAM systems in recent years and present a comprehensive summary of the components and underlying requirements. The multi-sensor fusion SLAM systems are composed of odometry and mapping and the contributions of multi-sensor configuration are ubiquitous in odometry. The research works in recent years have centered on tightly coupled systems and fusion frameworks have been dominated by filter-based and optimization-based approaches with the emergence of several learning-based fusion algorithms. Disparities of sensors lead to differences in feature extraction, residual representation, and the necessity of spatial and temporal calibration, which can be completed offline and online.

We categorize the algorithms according to the fused sensor types and identify that there are commonalities in their improvements, which primarily lie in refinements of fusion frameworks, feature extraction, dynamic processing, degradation detection, online calibration, and data association. Furthermore, we collate the public datasets frequently used for performance evaluation and propose several future challenges that a high-level long-term stable multi-sensor fusion SLAM system may encounter. We hope this paper will foster additional exploration into combining the complementary strengths of heterogeneous sensors.

# Declarations

Some journals require declarations to be submitted in a standardised format. Please check the Instructions for Authors of the journal to which you are submitting to see if you need to complete this section. If yes, your manuscript must contain the following sections under the heading 'Declarations':

- Funding
- Conflict of interest/Competing interests (check journal-specific guidelines for which heading to use)
- Ethics approval and consent to participate
- Consent for publication
- Data availability
- Materials availability
- Code availability
- Author contribution

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